

RAFAEL RICHIE

Fullstack Developer | AI/LLM Engineer

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PROFESSIONAL SUMMARY

Experienced Fullstack Developer with expertise in web application development, LLM-based chatbot engineering, agentic AI systems, and robotic process automation. Skilled in building end-to-end AI solutions using Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), and multi-agent orchestration frameworks.

WORK EXPERIENCE

PT. Prima Integrasi Network — Full Stack Developer

June 2024 – Present

- Developed LLM-powered chatbot using Streamlit Python and Ollama LLM for internal automation use cases
- Built agentic AI systems using Model Context Protocol (MCP) for advanced AI-driven process automation
- Engineered RAG (Retrieval-Augmented Generation) pipeline with Langchain, Ollama, ChromaDB, and VectorDB integrated with ReactJs frontend
- Developed social media super-app using ReactJs and Python Flask
- Implemented Robotic Process Automation (RPA) workflows using TagUI to streamline business processes
- Built and maintained applications across multiple databases: Couchbase, SQLite, MySQL, SinglestoreDB, MongoDB
- Implemented application security with JWT token authentication, 2FA via Google and Microsoft
- Developed responsive UI with Bootstrap, TailwindCSS, PrimeReact, and VantaJS
- Built conference meeting integration using Jitsi Meet
- Developed company profile websites and internal tools

PT Rakai Digital — Full Stack Developer (Freelance)

June 2025 – Present

- Developed Android transportation app using React Native featuring ticket booking system, real-time GPS tracking, payment gateway integration, and support for both public transit and private taxi services
- Built a tenant ticketing system for maintenance requests using React Native (Android), integrated with WhatsApp via Model Context Protocol (MCP) for automated notifications and workflow updates
- Developed a healthcare platform for nurses and doctors comprising a React Vite web app and React Native mobile app, with payment gateway and real-time online consultation chat (prototype)
- Developed a cooperative inter-store integration application to synchronize inventory and operations across multiple cooperative branches (prototype)

Youthopia, Malaysia — Full Stack Developer (Internship)

February 2024 – March 2024

- Developed company profile web application with ReactJs frontend and NodeJs backend server
- Integrated Stripe payment system using NodeJs for online transactions
- Implemented UI/UX design based on client requirements

RPG Ventures, Malaysia — Data Analyst & Frontend Developer (Apprenticeship)

November 2023 – December 2023

- Built internal applications using Google AppSheet
- Performed data cleansing and transformation using DBT (Data Build Tool) pipelines
- Explored and implemented Python automation scripts for operational workflows

EDUCATION

S1 Sistem Informasi — Universitas Pembangunan Jaya

September 2022 – Present

Bachelor's Degree in Software Engineering (BDSE) — Lithan Academy

September 2022 – October 2023

TECHNICAL SKILLS

AI / LLM / Agentic Systems

- LLM Development: Ollama LLM, Langchain, Streamlit Python
- Agentic AI: Model Context Protocol (MCP), MCP orchestrator, multi-agent workflow orchestration, tool-use integration
- RAG (Retrieval-Augmented Generation): ChromaDB, VectorDB, document ingestion pipelines
- AI Automation: Robotic Process Automation (RPA) with TagUI, process orchestration

Frontend Development

- ReactJS, React Vite, Typescript, NextJS
- React Native (mobile)
- HTML, CSS, JavaScript, JQuery
- Bootstrap, TailwindCSS, PrimeReact, VantaJS
- Axure (UI/UX prototyping)

Backend Development

- NodeJs, Flask Python, Java Spring Boot, Spring MVC

Databases

- Relational: MySQL, SQLite
- NoSQL: Couchbase, MongoDB, SinglestoreDB
- Vector: VectorDB, ChromaDB

Tools & Platforms

- Security: JWT Authentication, 2FA (Google & Microsoft)
- Payment: Stripe API
- Video Conferencing: Jitsi Meet integration
- Data: DBT Pipelines, Google AppSheet